

An exact exclusion of three-term zero sums among algebraic conjugates of degree 28

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Abstract

We prove that no irreducible polynomial of degree 28 over the rationals has three pairwise distinct roots whose sum is zero. The proof reduces the question to triple-incidence spans for minimally transitive permutation groups. Using Hulpke’s classification, we check all 30 such actions of degree 28. Their 1,025 triple orbits each contain 28 incidence vectors with a determinant nonzero modulo 101. The supplied exact certificates and a standard-library Python verifier make this finite calculation reproducible. This excludes the degree-28 case left undecided in the specified version of the work of Baronénas, Drungilas and Jankauskas.

1 Statement and context

Three algebraic conjugates here mean three roots of the same irreducible polynomial over $\mathbb{Q}$, viewed in a fixed algebraic closure $\overline{\mathbb{Q}}$. The roots in the question are required to be pairwise distinct.

Theorem 1. *There is no irreducible polynomial $f \in \mathbb{Q}[t]$ of degree 28 with pairwise distinct roots $\alpha, \beta, \gamma \in \overline{\mathbb{Q}}$ satisfying*

$$\alpha + \beta + \gamma = 0.$$

Baronénas, Drungilas and Jankauskas [1] study which degrees admit such a relation. Their Theorems 1 and 2 establish existence at degree 20 and several nonexistence results; Section 5 of version 1 explicitly lists degree 28 among the undecided degrees. Theorem 1 settles this degree by an exact finite calculation. The relation-space viewpoint is part of the existing approach to conjugate relations. We use Hulpke’s classification of minimally transitive permutation groups [3, 2], without claiming a further group classification. The contribution is the complete degree-28 exclusion and its explicit rank certificates.

The proof has two parts. A rational relation among roots gives relations on an orbit of three-element subsets under the Galois group. It suffices to consider minimally transitive subgroups, and every orbit of each such degree-28 action has full rational incidence rank. The latter assertion is verified by small integer minors. As adjacent computational context, Zeng [4] also uses exact finite linear-algebra certificates and a separate verifier, in the different setting of the weak Lefschetz property. That reference is methodological context only.

2 The permutation-group reduction

Let $\Omega = \{1, \dots, 28\}$ and let $e_1, \dots, e_{28}$ denote the standard basis of $\mathbb{Q}^{28}$. For $T \in \binom{\Omega}{3}$ and $H \leq S_{28}$, define

$$v_T = \sum_{i \in T} e_i, \quad W_{H,T} = \text{span}_{\mathbb{Q}}\{v_{hT} : h \in H\}. \quad (1)$$

A transitive group is *minimally transitive* if it has no proper transitive subgroup. Its action need not be regular.

Lemma 2. *Suppose that $W_{H,T} = \mathbb{Q}^{28}$ for every three-element subset T and every representative H of the minimally transitive degree-28 actions, up to conjugacy in S_{28} . Then $W_{G,T} = \mathbb{Q}^{28}$ for every transitive $G \leq S_{28}$ and every three-element subset T .*

Proof. Choose a transitive subgroup $K \leq G$ of least order. It is minimally transitive. Conjugation of K in S_{28} only permutes coordinates in (1) and permutes the three-element subsets. The hypothesis therefore gives $W_{K,T} = \mathbb{Q}^{28}$. Since $K \leq G$, we have $W_{K,T} \subseteq W_{G,T}$, proving the claim. $\square$

Lemma 3. *If $W_{G,T} = \mathbb{Q}^{28}$ for every transitive subgroup $G \leq S_{28}$ and every three-element subset T , then Theorem 1 holds.*

Proof. Suppose the prohibited polynomial exists, and let L be its splitting field over $\mathbb{Q}$. Characteristic zero implies that its 28 roots $\alpha_1, \dots, \alpha_{28}$ are distinct. Irreducibility implies that $G = \text{Gal}(L/\mathbb{Q})$ acts transitively on them, giving a subgroup of S_{28} . Define the $\mathbb{Q}$ -linear map

$$\Phi : \mathbb{Q}^{28} \longrightarrow L, \quad \Phi(e_i) = \alpha_i.$$

If the triple indexed by T sums to zero, then $\Phi(v_T) = 0$. Every $g \in G$ fixes $\mathbb{Q}$, so applying g to the equality gives $\Phi(v_{gT}) = 0$. Hence $W_{G,T} \subseteq \ker \Phi$. Full span forces $\Phi = 0$, so every root is zero, contradicting their distinctness. $\square$

This reduction imposes no restriction on solvability of the Galois group, reality of the roots, integrality or height. It is sufficient to exclude all transitive permutation actions, whether or not each action occurs as a Galois group over $\mathbb{Q}$.

3 The complete list of actions

We use Section 12.3 and Table 2 of the author-hosted version of Hulpke [3]. The associated data [2] give the following complete list of minimally transitive degree-28 actions, written as indices i for `TransitiveGroup(28,i)`:

$$\begin{aligned} \mathcal{I} = \{ & 1, 2, 3, 4, 11, 20, 32, 35, 42, 43, 48, 55, 56, 57, 58, \\ & 61, 66, 98, 105, 120, 152, 153, 200, 201, 262, 263, 371, \\ & 630, 795, 1169 \}. \end{aligned} \quad (2)$$

The first four actions are regular of order 28. The remaining 26 are the nonregular actions in Hulpke's table. Their two generators are printed on page 15 of the generator supplement. The permutations serialized in our certificate agree literally, in the same labeling and order, with those published generator pairs. This checks the correspondence of permutation actions, rather than merely comparing abstract group orders.

The four regular actions can also be checked directly. A group of order 28 has a unique Sylow 7-subgroup: its number divides 4 and is 1 modulo 7. Writing this subgroup as C_7 and taking a Sylow 2-subgroup P , we get $C_7 \rtimes P$, where P is C_4 or $C_2 \times C_2$. A homomorphism

$P \rightarrow \text{Aut}(C_7) \cong C_6$ has either trivial image or image of order two. For each of the two choices of P , all nontrivial choices are equivalent under automorphisms of P . There are thus at most four isomorphism types. Direct closure of the four serialized generating sets gives regular groups of order 28 with four different element-order distributions, so all four types are present. Isomorphic groups have equivalent regular actions under point relabeling.

The completeness of the 26 nonregular actions is the external classification input. We do not rederive it. Nor do we directly perform the triple-rank calculation on every one of the 1,854 transitive groups in the degree-28 library: Lemma 2 supplies the passage from the complete minimal list to all transitive groups.

4 Exact rank certificates

For an orbit $\mathcal{O} \subseteq \binom{\Omega}{3}$, let $M_{\mathcal{O}}$ be the matrix with rows v_T , $T \in \mathcal{O}$, and columns indexed by $1, \dots, 28$. Its rational row space is $W_{H,T}$ for any $T \in \mathcal{O}$.

Proposition 4 (Exact finite calculation). *For each action indexed by $\mathcal{I}$, every triple orbit contains 28 triples whose incidence matrix has determinant nonzero modulo 101. There are 1,025 such orbits in total. Consequently, $W_{H,T} = \mathbb{Q}^{28}$ for every action in the list and every three-element subset T .*

Certificate and verification. The supplementary file `rank-certificates.json` specifies the generators of each action as image arrays. For every triple orbit it gives the lexicographically least triple, the orbit size, an ordered list of 28 triples, the determinant of their incidence matrix over the integers, and its residue modulo 101. All 1,025 residues are nonzero. Table 1 records the complete coverage.

The verifier reconstructs the orbits on all $\binom{28}{3} = 3276$ triples by union-find. For each generator g it joins T to gT . The resulting connected components are exactly the group orbits: edges apply a generator or its inverse, and every group element is a word in those generators and inverses. The verifier requires exactly one certificate for each component, checks the stated orbit size, and checks membership of all 28 rows in that component.

For each chosen minor, modular Gaussian elimination recomputes its determinant in $\mathbb{F}_{101}$. Row swaps change the determinant sign; a nonzero pivot is used to eliminate the entries below it. The determinant is the signed product of the pivots. Every computed residue agrees with the stored residue and is nonzero. Since the incidence matrix has integer entries, its determinant over $\mathbb{Z}$ is nonzero as well. Thus its 28 rows are independent over $\mathbb{Q}$, and $W_{H,T} = \mathbb{Q}^{28}$.

The orbit sizes sum to 3,276 for each of the 30 actions, covering 98,280 pairs consisting of an action and a triple. No sampling is involved. $\square$

Proof of Theorem 1. The classification supplies the complete list in (2). Proposition 4 and Lemma 2 give full rational triple-incidence span for every transitive degree-28 group. Lemma 3 then excludes the required relation. $\square$

5 Reproduction and scope

The supplement contains the certificate, the published-generator transcription, the saved generator output, and a read-only verifier. From the manuscript directory, run

```
python3 supplement/verify-certificates.py
```

Table 1: Complete minimal-action coverage. For each action, the number of certified nonsingular minors equals the number of triple orbits.

Index	Order	Orbits	Index	Order	Orbits
1	28	117	61	448	19
2	28	117	66	448	26
3	28	117	98	896	40
4	28	117	105	896	45
11	56	65	120	1,092	5
20	112	49	152	1,344	8
32	168	29	153	1,344	10
35	196	39	200	2,184	5
42	336	15	201	2,184	5
43	336	15	262	3,584	20
48	392	26	263	3,584	20
55	392	26	371	10,752	7
56	392	17	630	76,832	14
57	392	17	795	172,032	7
58	392	17	1169	802,816	11
30 actions; 1,025 orbits; 1,025 certified rank-28 minors					

The program uses only the Python standard library. It does not call GAP, import the certificate producer, or download data. Run it with assertions enabled, as in the displayed command. In addition to the orbit and minor checks above, it validates the group index list, the permutation arrays, point transitivity, and literal agreement of the 26 nonregular generator pairs with the supplied transcription. The transcript reports 30 actions, 1,025 orbits, and 1,025 nonsingular minors modulo 101.

The initial computation used GAP 4.15.1 with TransGrp 3.6.5 to obtain the displayed actions. A separate Python producer reconstructed triple orbits by breadth-first closure, selected independent rows modulo 101, and computed integer determinants by fraction-free Bareiss elimination. The union-find verifier uses a different orbit construction and separate modular elimination. An additional verification pass reconstructed all orbits with integer bit masks and depth-first closure and recomputed all 1,025 integer determinants with SymPy 1.14.0. Their absolute values range from 3 to 331,776. Five deliberately corrupted inputs—a missing group, a missing orbit, repeated rows, a row from a different orbit, and an incorrect residue—were all rejected. The supplied verifier is sufficient to replay the modular certificates; the additional checks do not replace the classification input.

The mathematical statement is restricted to degree 28 over $\mathbb{Q}$ and to three distinct conjugates with coefficients all equal to one. We assert no extension to other degrees, coefficient patterns, or positive characteristic. The literature comparison is bounded to the sources described here and inspected on September 7, 2026; it does not establish exhaustive priority. In particular, our historical comparison is to version 1 of [1].

Computational and AI assistance. OpenAI Codex assisted with the proof organization, certificate-generation and verification programs, source comparison, and manuscript preparation. A separate model context performed an additional mathematical and exact computational review. These are disclosed computational assistance and review steps, not external peer review or statistically independent evidence. No proof-assistant formalization is claimed. The argument rests on the displayed mathematical reductions, the cited classification, and the reproducible exact certificates.

References

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